Values alignment review literature

A comprehensive annotated bibliography of review and survey articles produced, published, or materially updated from October 2025 through October 2026.

Coverage

12 included reviews across socio-technical process, advisory AI, agentic systems, pluralism, social choice, game theory, model internals, reward design, and deployment safety.

Reading aid

Each annotation identifies the review's scope, organizing method, main contribution, practical use, and publication status.

What changed in the review literature

> The past year's review literature moved the alignment conversation away from a single optimization problem and toward a layered socio-technical field: values are elicited, aggregated, encoded, evaluated, contested, and revised across the lifecycle of an AI system.

FOUR CONVERGING SHIFTS

From a target to a process. The strongest field-level review treats alignment as an ongoing sequence of choices involving designers, users, institutions, and adaptive systems. Advisory and agentic reviews carry that logic into concrete domains.

From one preference to many. Social-choice, game-theoretic, and pluralistic-alignment surveys reject the idea that disagreement is merely noisy supervision. They make representation, aggregation, incentives, and temporal change central research objects.

From output behavior to mechanisms. Reward-design and mechanistic-interpretability surveys examine the internal signals and representations through which values are learned and expressed.

From training to deployment. Safety-oriented reviews place alignment beside threat models, metrics, guardrails, and operational controls, emphasizing defense in depth.

HOW TO USE THIS BIBLIOGRAPHY

Begin with McKinlay et al. for a field map; use Triantafyllopoulos et al. or Zeng et al. for applied system contexts; use Halpern et al., Cai et al., or the pluralism survey for disagreement; and use Ji et al. or Naseem for model-level mechanisms.

The corpus is heterogeneous. "Comprehensive" and "systematic" are author claims with different standards. Publication status is therefore shown prominently on every entry.

Bottom line. The clearest cross-cutting gap is evaluation: the field still lacks widely accepted tests that connect stated values, internal mechanisms, multi-party disagreement, and long-horizon behavior under deployment conditions.

Inclusion rules and search boundary

> This is a bounded scholarly scan, not a claim of perfect bibliographic exhaustiveness. It captures publicly discoverable English-language review and survey articles available by 10 October 2026.

INCLUDED

- Review, systematic-review, or survey articles whose central subject is AI/LLM value alignment, human-intent alignment, pluralistic alignment, or a directly enabling alignment mechanism.
- Articles first published, accepted, posted, or materially revised within approximately 10 October 2025–10 October 2026.
- Peer-reviewed journal and conference papers, accepted manuscripts, and clearly labeled preprints or working papers.

INCLUDED WITH A BOUNDARY LABEL

- Broad safety reviews when alignment is an explicit organizing component.
- An older preprint with a substantial in-window revision, because the revised survey is part of the current literature but is not a newly created 2026 work.

EXCLUDED

- Empirical studies that administer surveys to people or models but do not review a body of literature.
- Position papers and research roadmaps without a review method, even when they synthesize prior work.
- General responsible-AI, fairness, interpretability, or LLM-safety surveys in which value alignment is incidental.
- Reviews published before the window without a clearly documented in-window version or venue publication.

Verification note. Dates, author lists, venues, sample sizes, and status labels were checked against publisher pages, institutional records, or repository metadata. Preprints can change; cite the version actually consulted.

Twelve reviews, seven vantage points

> No single article covers the full chain from human values to deployed behavior. Together, the corpus forms a layered map.

#	Review	Primary lens	Status class
01	**Alignment as process** Understanding the Process of Human-AI Value Alignment	Systematic review · 172 papers	Peer-reviewed journal article
02	**Advisory AI** The value alignment problem in advisory AI: a systematic literature review	PRISMA 2020 review · 83 studies (2011–2025)	Peer-reviewed journal article
03	**Friendly AI** Towards friendly AI: a comprehensive review and new perspectives on human-AI alignment	Comprehensive narrative review	Peer-reviewed journal article
04	**Agentic systems** Application-Driven Value Alignment in Agentic AI Systems: Survey and Perspectives	Application-driven taxonomy	Conference survey / proceedings version
05	**Social choice** AI Alignment From Social Choice Perspectives	Formal survey · social-choice theory	Accepted survey / preprint
06	**Game theory** AI Alignment through a Game-theoretic Lens: A Survey	Game-theoretic synthesis	Peer-reviewed conference paper / preprint
07	**Model internals** Mechanistic Interpretability for Large Language Model Alignment: Progress, Challenges, and Future Directions	Focused technical survey	Preprint
08	**Pluralistic alignment** Towards Pluralistic Alignment of LLMs: A Comprehensive Survey	Comprehensive survey · pluralism	Non-peer-reviewed review preprint
09	**Reward design** A Survey of Progress in LLM Alignment From the Perspective of Reward Design	Technical taxonomy · reward design	Peer-reviewed journal review
10	**Alignment and safety** Alignment and Safety in Large Language Models: Methods, Challenges, and Future Directions	Broad narrative survey	Working paper / SSRN preprint
11	**Operational pipeline** A Survey of LLM Alignment: Instruction Understanding, Intention Reasoning, and Reliable Generation	Task-oriented survey	Preprint
12	**Deployment safety** Survey on LLM Safety: Attacks, Defenses, Alignment, Metrics, and Guardrails	Broad safety survey with alignment coverage	Peer-reviewed journal survey

Reading sequence. For breadth: 01 → 04 → 08. For formal foundations: 05 → 06. For implementation: 09 → 07 → 12. For governance and application: 02 → 03 → 10.

Alignment as process

01

Understanding the Process of Human-AI Value Alignment

McKinlay, James; De Vos, Marina; Hoffmann, Manuel; and Theodorou, Andreas

Peer-reviewed journal article · March 2026

Systematic review · 172 papers

McKinlay, J., De Vos, M., Hoffmann, M., & Theodorou, A. (2026). Understanding the Process of Human-AI Value Alignment. Journal of Artificial Intelligence Research, 85, Article 29.

REVIEW FOCUS

A field-level review of how researchers frame, operationalize, and assess value alignment across human-AI systems. The paper moves beyond a narrow “target objective” view and asks what alignment work looks like as an ongoing process involving people, institutions, design choices, and adaptive systems.

HOW IT ORGANIZES THE FIELD

The authors systematically synthesize 172 papers and organize the evidence into six themes: motivations and approaches; practical and conceptual challenges; the values being aligned; relevant cognitive processes; human-agent teaming; and design. This gives the review unusual breadth across technical and human-centered traditions.

MAIN CONTRIBUTION

- Alignment is not one intervention but a chain of decisions about whose values count, how they are represented, and how conflict and change are handled.
- The literature is fragmented across disciplines; shared process models could connect formal methods with participatory and organizational practice.
- Human cognition and team dynamics matter because people interpret, negotiate, and sometimes adapt to AI behavior after deployment.

CRITICAL USE NOTE

Best starting point for a broad literature review or research agenda. Its process framing is especially useful for comparing otherwise incompatible approaches. Its breadth is a strength, but readers still need domain-specific reviews for concrete implementation detail.

VERIFIED SOURCE

https://www.jair.org/index.php/jair/article/view/18846

Advisory AI

02

The value alignment problem in advisory AI: a systematic literature review

Triantafyllopoulos, Andreas; Paxinou, Penelope; Tzanoulinou, Katerina; Verykios, Vassilios S.; and Kalles, Dimitris

Peer-reviewed journal article · published 9 February 2026

PRISMA 2020 review · 83 studies (2011–2025)

Triantafyllopoulos, A., Paxinou, P., Tzanoulinou, K., Verykios, V. S., & Kalles, D. (2026). The value alignment problem in advisory AI: a systematic literature review. AI and Ethics, 6, Article 147.

REVIEW FOCUS

A domain-focused review of systems that recommend, advise, or otherwise shape human decisions. It treats value alignment as a practical design problem: an advisory system must infer, represent, and respect values while operating amid uncertainty, competing priorities, and changing contexts.

HOW IT ORGANIZES THE FIELD

Using PRISMA 2020, the authors screen literature from 2011–2025 and retain 83 studies. The synthesis distinguishes four families of alignment approaches and maps them to the advisory lifecycle, making the review more methodologically explicit than many narrative surveys in the area.

MAIN CONTRIBUTION

- Advisory contexts make misalignment consequential because recommendations can redirect user choices without fully determining them.
- Value elicitation, preference learning, explanation, and user control should be evaluated together rather than as isolated components.
- The evidence base remains uneven: operational definitions and evaluation criteria vary, and longer-term interaction effects are under-tested.

CRITICAL USE NOTE

Use this review when studying recommender systems, decision support, or assistants. Its transparent screening method supports reproducibility. The domain focus limits coverage of autonomous agents and foundation-model internals, but improves practical specificity.

VERIFIED SOURCE

https://link.springer.com/article/10.1007/s43681-026-01015-4

Friendly AI

03

Towards friendly AI: a comprehensive review and new perspectives on human-AI alignment

Sun, Yu; Li, Chenyang; Alturki, Ahmed; Murthy, Divya; and Schuller, Björn W.

Peer-reviewed journal article · published 8 March 2026

Comprehensive narrative review

Sun, Y., Li, C., Alturki, A., Murthy, D., & Schuller, B. W. (2026). Towards friendly AI: a comprehensive review and new perspectives on human-AI alignment. AI and Ethics, 6, Article 193.

REVIEW FOCUS

A broad “friendly AI” synthesis that places value alignment beside explainability, privacy, fairness, and affective computing. The review is less concerned with one optimization recipe than with the collection of properties that make AI interactions trustworthy and socially acceptable.

HOW IT ORGANIZES THE FIELD

The paper integrates several mature responsible-AI literatures and proposes a cross-domain perspective on human-AI alignment. It functions as an umbrella review: readers see how familiar governance and interaction concerns become alignment requirements when systems act with greater autonomy and social presence.

MAIN CONTRIBUTION

- Alignment quality cannot be separated cleanly from whether users can understand, contest, and safely rely on system behavior.
- Affect and interaction design influence perceived alignment, but perceived friendliness is not proof of genuine value compatibility.
- Trade-offs among privacy, personalization, fairness, and utility should be treated as first-class design questions rather than post-hoc checks.

CRITICAL USE NOTE

Useful for multidisciplinary teams and policy audiences that need a bridge between technical alignment and responsible-AI practice. Because the scope is expansive, its categories are best used as a checklist and research map, not as a substitute for detailed technical evidence.

VERIFIED SOURCE

https://link.springer.com/article/10.1007/s43681-026-01070-x

Agentic systems

04

Application-Driven Value Alignment in Agentic AI Systems: Survey and Perspectives

Zeng, Y.; Zhu, Y.; Qin, Y.; Wu, Y.; Cheng, Y.; Zhang, Y.; Jin, Z.; Shen, Y.; Wang, W.; Zhong, L.; and Xiong, H.

Conference survey / proceedings version · 2026; preprint revised 7 August 2025

Application-driven taxonomy

Zeng, Y., Zhu, Y., Qin, Y., Wu, Y., Cheng, Y., Zhang, Y., Jin, Z., Shen, Y., Wang, W., Zhong, L., & Xiong, H. (2026). A Survey on Value Alignment in Agentic AI Systems. Proceedings of IJCAI 2026. Preprint title: Application-Driven Value Alignment in Agentic AI Systems: Survey and Perspectives.

REVIEW FOCUS

A survey tailored to agentic AI: systems that plan, use tools, coordinate, and take actions over extended horizons. The paper argues that value alignment must be analyzed at multiple scales because agent behavior is shaped by social rules, organizational context, and low-level action selection.

HOW IT ORGANIZES THE FIELD

The authors organize alignment into macro, meso, and micro levels. This architecture links societal and institutional values to coordination mechanisms and individual agent behavior, then maps application settings and open technical questions onto those levels.

MAIN CONTRIBUTION

- Agentic systems magnify alignment problems through persistence, tool access, delegation, and multi-agent interaction.
- Alignment objectives can conflict across levels: a locally obedient action may undermine team or societal goals.
- Evaluation must include trajectories and downstream effects, not only single-turn responses or static benchmarks.

CRITICAL USE NOTE

A strong entry point for work on autonomous agents, multi-agent systems, and tool-using assistants. The multi-level framework helps prevent category errors, although the application-driven organization is not a systematic evidence audit and should be complemented with empirical evaluations.

VERIFIED SOURCE

https://arxiv.org/html/2506.09656v1

Social choice

05

AI Alignment From Social Choice Perspectives

Halpern, Daniel; Micha, Evi; Procaccia, Ariel D.; Schiffer, Benjamin; Shapira, Itai; and Zhang, Junlin

Accepted survey / preprint · 19 June 2026

Formal survey · social-choice theory

Halpern, D., Micha, E., Procaccia, A. D., Schiffer, B., Shapira, I., & Zhang, J. (2026). AI Alignment From Social Choice Perspectives. arXiv:2606.21550. Accepted for ACM SIGecom Exchanges.

REVIEW FOCUS

A review of alignment problems that arise when feedback comes from many people with conflicting or incomparable preferences. It imports social-choice concepts to clarify what an aggregation procedure assumes, whom it advantages, and which desirable properties cannot all be satisfied at once.

HOW IT ORGANIZES THE FIELD

The paper surveys preference aggregation, collective decision-making, and feedback mechanisms through a computational social-choice lens. Rather than treating disagreement as label noise, it treats it as the central object that an alignment method must represent and resolve.

MAIN CONTRIBUTION

- Aggregating human feedback always encodes normative choices, even when the mechanism appears purely statistical.
- Classical impossibility results and incentive concerns help explain why no single reward or preference model can satisfy every fairness and consistency criterion.
- Mechanism design suggests ways to reason about strategic reporting, representation, and accountability in feedback collection.

CRITICAL USE NOTE

Essential for pluralistic or democratic alignment projects. The formal lens makes assumptions explicit and supports precise comparisons. Its abstraction may feel distant from deployment practice; readers should connect the theory to concrete data-collection and governance procedures.

VERIFIED SOURCE

https://arxiv.org/abs/2606.21550v1

Game theory

06

AI Alignment through a Game-theoretic Lens: A Survey

Cai, Yanan; Zhao, Zhongrui; Lu, Zhigang; Lee, Ickjai; Zhang, Wei Emma; Xue, Minhui; Zhang, Yihong; Pang, Shuchao; and Xiang, Wei

Peer-reviewed conference paper / preprint · 28 August 2026

Game-theoretic synthesis

Cai, Y., Zhao, Z., Lu, Z., Lee, I., Zhang, W. E., Xue, M., Zhang, Y., Pang, S., & Xiang, W. (2026). AI Alignment through a Game-theoretic Lens: A Survey. arXiv:2608.27910. EMNLP 2026 main conference paper.

REVIEW FOCUS

A survey of alignment as strategic interaction among humans, models, institutions, and other agents. It asks when game-theoretic structure adds explanatory or design value, especially where preferences are contextual, non-transitive, or change over time.

HOW IT ORGANIZES THE FIELD

The literature is organized around three recurring challenges: preference diversity, alignment priority, and temporal dynamics. The review relates methods to game elements such as players, utilities, information, equilibrium behavior, repeated interaction, and commitment.

MAIN CONTRIBUTION

- Static preference fitting is poorly matched to environments where parties adapt, bargain, or manipulate signals.
- Priority conflicts—whose objectives dominate and when—can be modeled as institutional or mechanism choices, not merely optimization weights.
- Game theory is most useful when strategic assumptions are explicit; loose game metaphors can obscure rather than sharpen analysis.

CRITICAL USE NOTE

Particularly relevant to multi-agent alignment, human-AI bargaining, adversarial feedback, and governance. The survey complements social choice: social choice emphasizes aggregation, while game theory emphasizes incentives and dynamics.

VERIFIED SOURCE

https://arxiv.org/abs/2608.27910

Model internals

07

Mechanistic Interpretability for Large Language Model Alignment: Progress, Challenges, and Future Directions

Naseem, Usman

Preprint · 21 January 2026
Focused technical survey

Naseem, U. (2026). Mechanistic Interpretability for Large Language Model Alignment: Progress, Challenges, and Future Directions. arXiv:2602.11180.

REVIEW FOCUS

A technical survey of how mechanistic interpretability can support alignment by exposing internal computations rather than relying only on observed outputs. It connects circuit analysis and representation-level interventions to familiar alignment methods and oversight challenges.

HOW IT ORGANIZES THE FIELD

The review spans circuit discovery, feature visualization, activation steering, and causal intervention, then considers their relationship to RLHF, constitutional approaches, and scalable oversight. It frames interpretability both as diagnosis and as a possible control surface.

MAIN CONTRIBUTION

- Internal evidence can reveal failure modes that behavior-only tests miss, but current explanations are often partial and model-specific.
- Superposition, polysemantic features, scale, and emergent behavior complicate stable interpretation.
- Automated methods, cross-model circuit transfer, and causal validation are necessary before interpretability can bear major safety claims.

CRITICAL USE NOTE

Best for readers investigating model-internal alignment assurance. It is valuable as a compact map of techniques, but should not be read as evidence that interpretability has already delivered reliable certification; most capabilities remain research-stage.

VERIFIED SOURCE

https://arxiv.org/abs/2602.11180v1

Pluralistic alignment

08

Towards Pluralistic Alignment of LLMs: A Comprehensive Survey

Naseem, Usman, et al.

Non-peer-reviewed review preprint · 2026

Comprehensive survey · pluralism

Naseem, U., et al. (2026). Towards Pluralistic Alignment of LLMs: A Comprehensive Survey. Preprints.org manuscript 202603.1876.

REVIEW FOCUS

A dedicated survey of methods that aim to represent diverse values rather than compress them into one average target. It extends pluralistic alignment beyond text-only language models and connects methods, datasets, metrics, and underexplored application settings.

HOW IT ORGANIZES THE FIELD

The paper organizes the field around three objectives: Overton pluralism (covering a reasonable range), steerable pluralism (conditioning on a perspective), and distributional pluralism (matching a population distribution). It then maps training, inference, data, and evaluation approaches to those modes.

MAIN CONTRIBUTION

- The three pluralism modes are not interchangeable and require different datasets and evaluation criteria.
- A model may be steerable without covering minority views, or broad in coverage without matching any real population.
- Open problems include multi-turn consistency, temporal and geographic change, agentic settings, model internals, and risks created by malicious or unsafe perspectives.

CRITICAL USE NOTE

The most targeted recent map of pluralistic-alignment techniques. Treat its preprint status and evolving byline as important metadata; verify the version used before formal citation. Its taxonomy is useful for avoiding vague claims that a system is simply “pluralistic.”

VERIFIED SOURCE

https://www.preprints.org/manuscript/202603.1876

Reward design

09

A Survey of Progress in LLM Alignment From the Perspective of Reward Design

Ji, Miaomiao; Wu, Yanqiu; Wu, Zhibin; Wang, Shoujin; Yang, Jian; Dras, Mark; and Naseem, Usman

Peer-reviewed journal review · early online 23 January 2026; issue July 2026

Technical taxonomy · reward design

Ji, M., Wu, Y., Wu, Z., Wang, S., Yang, J., Dras, M., & Naseem, U. (2026). A Survey of Progress in LLM Alignment From the Perspective of Reward Design. IEEE Transactions on Artificial Intelligence, 7(7), 3688–3702. DOI: 10.1109/TAI.2026.3654921.

REVIEW FOCUS

A structured account of rewards as the bridge between human or rule-based feedback and model optimization. The survey treats reward design broadly: mathematical form, construction pipeline, role in training or inference, and evaluation.

HOW IT ORGANIZES THE FIELD

The review asks how rewards are specified, how they are built from rule-based, data-driven, or hybrid supervision, how they interact with RLHF, direct preference optimization, and in-context methods, and how they are evaluated for robustness and downstream impact.

MAIN CONTRIBUTION

- Reward design choices determine what information is preserved or collapsed before optimization begins.
- The field is moving toward hybrid supervision, multidimensional signals, and methods that do not require a conventional reinforcement-learning loop.
- Evaluation needs to test robustness, generalization, safety diagnostics, and downstream behavior rather than reward-model accuracy alone.

CRITICAL USE NOTE

A central reference for post-training research. Its taxonomy supports comparison across methods that are often described with incompatible terminology. The reward-centered framing is powerful but does not resolve the prior normative question of which values or stakeholders the reward should represent.

VERIFIED SOURCE

https://researchers.mq.edu.au/en/publications/a-survey-of-progress-in-llm-alignment-from-the-perspective-of-rew/

Alignment and safety

10

Alignment and Safety in Large Language Models: Methods, Challenges, and Future Directions

Ding, Yu; Han, Xudong; Yang, Junjie; Wang, Tianyang; Bi, Ziqian; Song, Xinyuan; Hao, Junfeng; Song, Junhao; Ge, Enze; Peng, Benji; Liu, Ziwei; Liang, Chia Xin; Zhang, Yichao; Liu, Ming; Xu, Jiawei; Huang, Binhua; Yu, Zhenyu; Mo, Yang; Qiao, Jing; Zhang, Danyang; and Ma, Yue

Working paper / SSRN preprint · posted 27 January 2026

Broad narrative survey

Ding, Y., Han, X., Yang, J., et al. (2026). Alignment and Safety in Large Language Models: Methods, Challenges, and Future Directions. SSRN Electronic Journal. DOI: 10.2139/ssrn.6008914.

REVIEW FOCUS

A broad survey connecting technical alignment and safety techniques with ethical-value concerns. It covers training and adaptation methods, benchmarks, localized alignment, and the tension between safety interventions and other model qualities.

HOW IT ORGANIZES THE FIELD

The paper synthesizes work on methods including human-feedback training and adapter-based interventions, then discusses ethical safety, regional variation, evaluation, and a forward-looking roadmap. It is a narrative overview rather than a reproducible systematic review.

MAIN CONTRIBUTION

- Safety and alignment evaluations need to account for social and regional context, not only universal benchmark averages.
- Interventions can create trade-offs among creativity, fairness, risk aversion, and helpfulness.
- Localized evaluation is presented as a practical requirement for models intended to reflect diverse ethical standards.

CRITICAL USE NOTE

Useful as an orientation document and as evidence that the safety and values literatures are converging. Because it is a working paper with a very broad scope, use it to discover themes and references rather than as the sole authority for comparative claims.

VERIFIED SOURCE

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6008914

Operational pipeline

11

A Survey of LLM Alignment: Instruction Understanding, Intention Reasoning, and Reliable Generation

Chang, Zongyu; Lu, Feihong; Zhu, Ziqin; Li, Qian; Ji, Cheng; Yang, Tao; Chen, Zhuo; Peng, Hao; Liu, Yang; Xu, Ruifeng; Song, Yangqiu; Li, Jianxin; and Wang, Shangguang

Preprint · original 2025; substantially updated 29 January 2026

Task-oriented survey

> Chang, Z., Lu, F., Zhu, Z., Li, Q., Ji, C., Yang, T., et al. (2026 revision). A Survey of LLM Alignment: Instruction Understanding, Intention Reasoning, and Reliable Generation. arXiv:2502.09101.

REVIEW FOCUS

A task-oriented survey that operationalizes alignment through three stages: understanding instructions, reasoning about user intentions, and generating reliable responses. It emphasizes ambiguity, long-context interaction, inconsistent commands, misinformation, and unethical outputs.

HOW IT ORGANIZES THE FIELD

The authors classify failure scenarios, compare solution families, and organize benchmarks around the three stages. This creates a practical pipeline from what the user says, through what the model infers, to what it ultimately produces.

MAIN CONTRIBUTION

- Alignment failures can originate before generation—through incorrect parsing or intention inference—so output filtering alone is insufficient.
- Long and multi-turn contexts create compounding ambiguity and reliability problems.
- Benchmarks should separate comprehension, reasoning, and generation failures so improvements are attributable.

CRITICAL USE NOTE

Included as an in-window revision rather than a newly created 2026 paper. It is useful for operational alignment and human-intent fidelity, but its treatment of plural values and collective disagreement is less developed than the dedicated reviews above.

VERIFIED SOURCE

http://arxiv.org/abs/2502.09101

Deployment safety

12

Survey on LLM Safety: Attacks, Defenses, Alignment, Metrics, and Guardrails

Jalan, Pratik; Abishethvarman, Vadivel; Chandna, Bhavik; and Naseem, Usman

Peer-reviewed journal survey · June 2026

Broad safety survey with alignment coverage

Jalan, P., Abishethvarman, V., Chandna, B., & Naseem, U. (2026). Survey on LLM Safety: Attacks, Defenses, Alignment, Metrics, and Guardrails. Machine Learning, 115(6), Article 130. DOI: 10.1007/s10994-026-07060-8.

REVIEW FOCUS

A broad safety survey that treats alignment as one component in a deployment stack alongside attacks, defenses, metrics, and guardrails. Its relevance here is not that every safety mechanism is value alignment, but that real systems combine normative objectives with adversarial and operational controls.

HOW IT ORGANIZES THE FIELD

The review spans threat models, mitigation strategies, alignment methods, evaluation, and runtime safeguards. It provides a systems view of how post-training and external controls interact across the lifecycle.

MAIN CONTRIBUTION

- Alignment training and deployment-time guardrails address different failure channels and should be evaluated jointly.
- Metrics can give a false sense of security if they do not match realistic attacks or user contexts.
- Defense-in-depth is necessary because no single post-training method reliably covers adversarial and distributional shifts.

CRITICAL USE NOTE

Included as adjacent-but-relevant literature because it places alignment inside a full safety architecture. Readers focused strictly on value elicitation or aggregation may treat it as secondary; deployment teams will find the integration perspective important.

VERIFIED SOURCE

DOI identifier in citation; publisher URL not reproduced because it was not independently resolved in this scan.

Where the reviews agree—and where they do not

> The corpus converges on the inadequacy of one-dimensional alignment scores, but it does not converge on a single replacement.

SHARED DIAGNOSIS

Values are plural, contextual, and dynamic. Feedback is filtered through data collection, aggregation, reward design, and institutional choices. Evaluation that ignores these layers can overstate progress.

SHARED DESIGN IMPLICATION

Alignment should be treated as an end-to-end lifecycle with explicit stakeholder assumptions, traceable objectives, contestability, and monitoring after deployment.

FAULT LINE: REPRESENTATION

Pluralistic and social-choice work asks whether systems should cover viewpoints, obey a selected viewpoint, or match a population distribution. Each answer creates different fairness and safety risks.

FAULT LINE: CONTROL SURFACE

Reward-design work intervenes through learning signals; interpretability work targets internal mechanisms; safety work layers external guardrails. None yet supplies complete assurance alone.

FAULT LINE: UNIT OF ANALYSIS

Some reviews focus on single-model responses, others on advisory relationships, long-horizon agents, teams, or institutions. Results do not automatically transfer across these levels.

EVIDENCE GAP

Only two entries use clearly reported systematic-review procedures and counts. Most surveys offer valuable taxonomies but limited evidence about search completeness, study quality, or effect size.

RESEARCH PRIORITIES SUGGESTED BY THE COMBINED CORPUS

- Evaluation suites linking stakeholder values to trajectories and downstream outcomes.
- Methods for representing disagreement without erasing minorities or enabling harmful personalization.
- Longitudinal tests for changing preferences, strategic adaptation, and institutional drift.
- Causal validation of internal alignment mechanisms across models and scales.
- Transparent reporting of who supplied feedback, how it was aggregated, and what trade-offs were chosen.

Important near-misses and why they are not core entries

Before window	**A Survey on Personalized and Pluralistic Preference Alignment in Large Language Models** (Xie et al.) was published in July 2025. It is highly relevant, but predates the 12-month boundary used here.
Before window	**Large Language Model Alignment: A Survey** and **AI Alignment: A Comprehensive Survey** are foundational background works, but their original publication dates fall outside the period.
Position paper	**A Roadmap to Impactful Pluralistic Alignment Research** sets a research agenda rather than conducting a literature review. It is useful context, not a review article under this bibliography's rule.
Empirical survey	**Pluralistic AI Alignment: A Cross-Cultural Pilot Survey** surveys participants. "Survey" here names the research method, not a review of prior literature.
Empirical studies	Recent work on cultural value measurement, model homogenization, Overton benchmarks, and user perceptions was excluded when its main contribution was a dataset, experiment, or benchmark rather than a literature synthesis.
Broad adjacency	General reviews of responsible AI, explainability, fairness, security, and LLM safety were excluded unless value alignment was an explicit central axis. Entry 12 is retained—and labeled—because alignment is named as a major section in its title and systems framing.

> **Interpretive consequence.** "All" should be read as all items found that satisfy these documented rules, not all scholarship that might use the word alignment. The boundary is necessary because "alignment" spans technical, moral, human-factors, and security literatures with inconsistent vocabulary.

Source register (1 of 2)

> Links point to publisher, institutional, or repository records used to verify bibliographic facts and review scope.

01. Understanding the Process of Human-AI Value Alignment
https://www.jair.org/index.php/jair/article/view/18846

02. The value alignment problem in advisory AI: a systematic literature review
https://link.springer.com/article/10.1007/s43681-026-01015-4

03. Towards friendly AI: a comprehensive review and new perspectives on human-AI alignment
https://link.springer.com/article/10.1007/s43681-026-01070-x

04. Application-Driven Value Alignment in Agentic AI Systems: Survey and Perspectives
https://arxiv.org/html/2506.09656v1

05. AI Alignment From Social Choice Perspectives
https://arxiv.org/abs/2606.21550v1

06. AI Alignment through a Game-theoretic Lens: A Survey
https://arxiv.org/abs/2608.27910

Publication labels describe status visible during this scan. Repository records and forthcoming conference metadata may be revised after 10 October 2026.

Source register (2 of 2)

> Links point to publisher, institutional, or repository records used to verify bibliographic facts and review scope.

07. Mechanistic Interpretability for Large Language Model Alignment: Progress, Challenges, and Future Directions

https://arxiv.org/abs/2602.11180v1

08. Towards Pluralistic Alignment of LLMs: A Comprehensive Survey

https://www.preprints.org/manuscript/202603.1876

09. A Survey of Progress in LLM Alignment From the Perspective of Reward Design

https://researchers.mq.edu.au/en/publications/a-survey-of-progress-in-llm-alignment-from-the-perspective-of-rew/

10. Alignment and Safety in Large Language Models: Methods, Challenges, and Future Directions

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6008914

11. A Survey of LLM Alignment: Instruction Understanding, Intention Reasoning, and Reliable Generation

http://arxiv.org/abs/2502.09101

12. Survey on LLM Safety: Attacks, Defenses, Alignment, Metrics, and Guardrails

Jalan, P., Abishethvarman, V., Chandna, B., & Naseem, U. (2026). Survey on LLM Safety: Attacks, Defenses, Alignment, Metrics, and Guardrails. Machine Learning, 115(6), Article 130. DOI: 10.1007/s10994-026-07060-8.

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